# Flame Extinguishment Experiment (FLEX)




The Flame Extinguishment Experiment (FLEX) experiment was designed to assess and quantify the effectiveness of inert-gas suppressants in microgravity and obtain the most conservative estimate of the limiting oxygen index for steady combustion. FLEX is studying the behavior of near-limit diffusion flames examining in detail liquid- and gas-phase transport and chemical kinetics and developed and is validating detailed and reduced-order transport and chemistry models that are the foundation for real engine simulations.

Flame Extinguishment Experiment (FLEX) will map the flammability boundaries for liquid fuel combustion in reduced gravity to quantify the suppressant efficacy of various gaseous suppressants over the range of candidate atmospheric pressures and oxygen concentrations. This investigation will develop predictive theoretical and numerical codes and chemical kinetic schemes to model flammability boundaries as a function effective gravitational acceleration on the unique ambient conditions encountered in space exploration applications. The development and validation of these models will require detailed spatially and temporally resolved measurements of droplet burning rate, flame extinction, flame radiation, soot concentration, soot temperature, etc. This investigation will also develop improved and validated reduced (simplified) theoretical and numerical sub-models of important physical processes (chemical kinetics, radiation, soot formation/destruction) that can be used in simulations of large scale, realistic fires.

The independent experiment variables analyzed by MDCA-FLEX are:

**Oxygen Mole Fraction:** The ambient oxygen mole fraction in a typical space environment can vary from high concentrations in EVA pre-breathing environments down to that typically found in air. At high oxygen concentrations, however, the chemical times are small enough (relative to the characteristic flow times) such that the droplets will burn to completion rather than exhibiting flame extinction. Therefore, it is also necessary to study low oxygen concentrations, down to the Limiting Oxygen Index (LOI). It is also important to determine the LOI to verify the chemical mechanisms. The oxygen mole fractions in the present study will vary from 0.10 to 0.50. The chamber must also be large enough such that there is no significant decrease in ambient oxygen mole fraction during an experiment (i.e. the droplet burns in an essentially constant ambient).

**Diluent (non-suppressant):** The ambient mixture will consist of oxygen mixed with a suppressant and the balance of an inert diluent gas. The diluent gas for these studies will be primarily nitrogen since that is the typical diluent on Earth and expected in space. There will be a small number of tests with a helium diluent gas. The reason is primarily for baseline comparisons with the Droplet Combustion Experiment, which used helium as the diluent gas and to vary the physical properties of the diluent gas.

**Suppressant Type:** The tests will examine candidate gaseous suppressants that have widely varying physical, chemical and radiative properties. This will enable model and sub-model development and validation over a wide range of ambient conditions to improve the predictive capabilities of the models. The suppressants are carbon dioxide and helium. The expected

concentrations range from 0.00 mole fraction to the limit where no flame can exist (0.70 expected for the least active suppressant).

**Pressure:** Ambient pressure does not significantly influence the droplet burning rate but does influence chemistry at sufficiently low values. In addition, one strategy for extinguishing a fire is to isolate the habitat where it exists and vent the cabin to space. It is therefore beneficial to have verified suppressant data at low pressures. The pressure for the tests will range from 0.5 to 1.0 atm (standard atmosphere). As with the oxygen mole fraction, it is important that the test chamber be sufficiently large so that the pressure is essentially constant during an experiment (i.e. the droplet burns in a constant pressure ambient).

**Fuel Type:** The advantage of the droplet geometry is that the fuel is relatively simple and better characterized than typical fuels in fire safety studies (e.g., PMMA or paper). This is also a disadvantage since it does not represent a practical fuel. The study will use two typical hydrocarbon fuels, an alcohol, methanol (CH3OH) and an alkane, heptane (C7H16). There is a relatively large experience base with these fuels. Methanol has a fuel-bound oxygen atom and burns with a very dim blue flame (not much soot production) with a small standoff distance, so it has widely different radiative characteristics than heptane. Therefore, studying these two fuels gives a wide range of fire scenarios to verify model and sub-model performance over.

**Droplet Diameter:** The droplet is ignited in the flammable region and then burns in a constant oxygen mole fraction ambient until flame extinction. Under the assumption of quasi-steady burning, the initial droplet size should not significantly influence the determination of the extinction droplet size. Transient influences, however, will be present, so some variation in initial droplet diameter is necessary to determine the deviation from quasi-steady behavior. The initial droplet size in the proposed study will vary between 2 and 5 mm.

**Related Documents**

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**Publications**

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# Flame Extinguishment Experiment-2 (FLEX-2)




The second in the FLEX series of experiments, the FLEX-2 investigation uses fuels and environmental conditions that mimic real combustor conditions. The investigation will extend and advance the research into droplet combustion, studying the influence of sub-buoyant convective flows on combustion rates, determining the influence of a second burning droplet on a linear array, and beginning the study of practical fuels by burning bi-component and surrogate fuels. As the research extends into increasingly complex fuels, FLEX-2 data can help verify models of real fuels used in transportation and industry. Results of the FLEX-2 experimental data will help to develop verified detailed and reduced-order models of droplet combustion, particularly with flow-field and droplet-droplet interactions.

The spherically symmetric combustion of a liquid fuel droplet in a quiescent oxidizing atmosphere is a classical problem in combustion research. The advantage of the spherical symmetry is that only one spatial dimension enters the description of the combustion process. This greatly simplifies the mathematics required for modeling combustion processes. Microgravity experiments provide the opportunity to achieve this spherical symmetry, a fact which NASA has used to its advantage in fundamental investigations for a number of years. The Fundamental Studies in Droplet Combustion and Flame Extinguishment in Microgravity (FLEX-2) investigation touches on a number of far-reaching combustion phenomena that can be uniquely addressed through the idealized geometry of a droplet burning in microgravity.

Studies of the liquid-phase mixing of the different components in fuels clarifies the way practical fuels turn to gas from liquid in the combustion process. Understanding soot formation during the combustion of liquid fuels is also studied. These studies are important due to soot's negative effects on the environment and on human health. Knowledge of soot formation also has practical uses in certain industrial processes. The interactions of droplets with their gaseous environments and with other droplets are studied to reveal the fundamental dynamics of combustion.

In the absence of gravity, small droplets of fuel (i.e., from 2 mm to 4 mm in diameter) burn "one-dimensionally", which means the flame will be shaped a like ball about the size of a large olive that will be centered around the droplet. This one-dimensional nature of the droplet flame allows the science team to easily measure and understand important features of the burning fuel that would otherwise be impossible to obtain on the ground. This particular type of flame configuration allows measurement and observation of very complex interactions in a spherically one-dimensional system, providing insights into the behavior of combustion phenomena that would otherwise be difficult, if not impossible, to obtain in multi-dimensional systems that are typically found in most 1-g fires.

The Fundamental Studies in Droplet Combustion and Flame Extinguishment in Microgravity – 2 (FLEX-2) experiments involve dispensing, deploying and igniting either a single droplet or a pair of droplets, depending on the experiment configuration, in a known and controlled gaseous environment. The ignited droplets either burn to completion or the flame extinguishes at a finite

droplet radius in a quiescent ambient. During the experiments involving binary droplet arrays, two droplets are placed along a support fiber at known locations and the droplets are ignited simultaneously and the subsequent burning is recorded. In the convective flow tests, a single droplet is subjected to a known and controlled sub-buoyant flow of the quiescent atmosphere around the droplet. The flow is realized when the droplet is tethered to a substrate. In this tethered mode the droplet is placed at the intersection of two fiber crosshairs. The cross hairs are then moved through the quiescent atmosphere surrounding the droplet.

The experiments in FLEX-2 can easily be categorized into four separate groupings; these being, the pure fuel experiments, the fuel mixture experiments, the convective flow experiments, and the binary droplet array experiments. Each of these groupings are sponsored by one or more investigators comprising the investigation team and entail anywhere from 30 to 60 separate tests using different fuels, different ambient conditions, or different droplet configurations.

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## Flame Extinguishment Experiment -2 JAXA (FLEX-2J)




The FLEX-2J experiment is a joint effort between NASA and the Japanese Space Agency, JAXA, as well as Nihon University and Yamaguchi University. Derived from the JAXA Group Combustion Experiment science objectives, the FLEX-2J will complement those goals using the NASA FLEX-2 hardware and combustion facilities on ISS. FLEX-2J will observe and measure fuel droplet motions during flame spreading along a one-dimensional droplet array. Three droplets will be deployed to fixed positions upon ceramic beads on a silicon carbide fiber. Then an additional three to ten movable droplets are positioned to the fiber at known locations. The first fixed droplet is ignited, and the flame is propagated down the array from droplet to droplet. The subsequent burning and motions of the unpinned droplets are recorded; particularly the velocities of the free droplets before and after flame spread are measured. In addition, the experiment will obtain the history of flame leading edge position, flame spread limit span, and the growth process of the group flame along the fuel droplet array. Specifically, the

experiment will measure burning rate, burning time, flame spread and droplet motion as a function of inter-droplet spacing, ambient pressure and gas composition.

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## FLEX-Italian Combustion Experiment for Green Air (FLEX-ICE-GA)




The FLEX-Italian Combustion Experiment for Green Air will test surrogate fuels as defined by the Italian Space Agency (ASI) within the CIR in the FLEX-2 configuration. A collaborative agreement between U.S. and Italian scientists from the Italian National Research Council–Istituto Motori will allow collaboration on research into biologically derived fuels (biofuels) in an investigation into new, green energy sources. Researchers from the NRC–Istituto Motori have identified the fuels to be used as 50–50 mixtures of n-heptane/ethanol and 50–50 n-hexanol/n-decane.

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